AI in Healthcare – SAT5114 | Group 4

Magseed CNN

A Machine Learning Approach to Radiological Guide Placement Optimization

ResUNet with Attention Gates · CBIS-DDSM Dataset · IoU = 0.9216

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Problem & Motivation

Clinical Context

170,000
lumpectomies per year
in the United States

~10 mm
required Magseed placement
accuracy from lesion center

30 days
Magseed can be placed
before surgery

The Challenge

- Nonpalpable tumors require imaging-guided localization before surgery
- Traditional wire localization is uncomfortable and limits scheduling flexibility
- Magseed improves flexibility but precise placement remains critical
- Incorrect placement can force repeat surgery and extended patient stress
- Radiologist workload and variability can affect placement accuracy

Proposed Solution

A deep learning project that detects breast lesions from mammography images and recommends optimal Magseed placement coordinates.

1	2	3	4	5	6
Input	**Encode**	**Attend**	**Decode**	**Threshold**	**Place**
Cropped ROI mammogram (grayscale, 384×384)	ResNet34 extracts features at 5 resolution scales	Attention gates suppress irrelevant background tissue	Decoder rebuilds full-res lesion probability map	Optimal threshold (0.56) binarizes the mask	Centroid → physical mm coordinate for Magseed

8-way Test Time Augmentation applied at inference for improved accuracy

Model Architecture — ResUNet with Attention Gates

ENCODER (ResNet34)		DECODER (with Attention)
Conv block 64ch · 192×192	- - - - →	**Up block 1** 256ch · 24×24
Res layer 1 64ch · 96×96	- - - - →	**Up block 2** 128ch · 48×48
Res layer 2 128ch · 48×48	- - - - →	**Up block 3** 64ch · 96×96
Res layer 3 256ch · 24×24	⟶ - - - - →	**Up block 4** 32ch · 192×192
Res layer 4 ★ 512ch · 12×12		**Output head** 1ch ⟶ Sigmoid

Key Design Decisions

Pretrained encoder

ResNet34 ImageNet weights transferred to mammography creates a faster convergence on limited data

Grayscale adaptation

First conv layer averaged across 3 channels to accept single-channel X-ray input

Attention gates

Each skip connection filtered by a learned attention map — suppresses irrelevant tissue

Dropout 0.2

Applied between conv layers in each decoder block to prevent overfitting

Differential LR

Encoder trained at 1e-5 (10× lower) to preserve pretrained features

Mixed precision

torch.amp used on A100 GPU for faster training without accuracy loss

Dataset — CBIS-DDSM

10,000+	2,863	704	80 / 20	Patient
Mammography images total	Training pairs used	Test pairs used	Train / test split ratio	Level split (no leakage)

⚠️ **Critical Implementation Note**

Model uses CROPPED ROI images paired with ROI masks — not full mammograms. Full mammograms paired with small ROI masks are spatially misaligned, which degraded IoU from 0.90 to 0.19 in earlier experiments. Alignment was verified by checking that image and mask dimensions match within 5 pixels.

Abnormality Types in Dataset

- **Mass cases:** Irregular, spiculated, lobular, and round
- **Calcification cases:** Fine pleomorphic, amorphous clusters
- **Pathology labels:** Benign, benign without callback, malignant
- **Annotations:** Bounding boxes + ROI segmentation masks

Training Augmentation

- Horizontal & vertical flip
- Rotation ±20°
- Brightness & contrast jitter
- Elastic transform
- Grid distortion
- Gaussian noise

Training Configuration & Evaluation Metrics

Training Configuration

Loss function	BCE + Dice (α = 0.5)
Optimizer	AdamW
Encoder LR	1e-5 (pretrained, fine-tune)
Decoder LR	1e-4 (randomly initialized)
Weight decay	5e-4
LR schedule	Cosine annealing
Epochs	40
Batch size	24 (A100 GPU)
Mixed precision	torch.amp (float16)
Dropout	0.2 in all decoder blocks

Evaluation Metrics

IoU (0.9216)

Intersection over Union — spatial overlap between predicted and true lesion. Directly determines placement accuracy.

Recall (0.9776)

% of true lesion pixels found. *Most critical clinically* Missing lesion tissue risks incomplete tumor removal.

Precision (0.9415)

% of predictions that are correct. High precision = tight boundary, less over-prediction into healthy tissue.

F1 (0.9592)

Harmonic mean of precision and recall. Overall balance of segmentation quality across the full test set.

Optimal threshold of 0.56 found via grid search across 0.30–0.55 on the validation set. 8-way TTA (4 rotations × 2 flips) applied at inference — averages 8 probability maps before thresholding.

Results — Model Evolution

Version	IoU	F1	Precision	Recall	Key Change
U-Net (CPU baseline)	0.430	0.588	0.575	0.655	Plain U-Net, full mammograms
ResUNet v4 (256px)	0.905	0.950	0.935	0.965	Pretrained ResNet34, aligned crops
ResUNet v5 (384px)	0.916	0.956	0.952	0.961	+ Attention gates, higher resolution
ResUNet v6 (final)	**0.922**	**0.959**	**0.942**	**0.978**	**+ GridDistortion, threshold=0.56, 8-way TTA**

+114%
IoU improvement
U-Net → v6

+1.1%
IoU gain
v4 → v6

0.56
Optimal
threshold

×8
TTA
predictions

Conclusion

What Was Achieved

✓ ResUNet + attention gates trained on 2,863 aligned mammogram/mask pairs

✓ Best IoU of 0.9216 — 114% improvement over plain U-Net baseline

✓ Recall of 97.8% — model finds nearly all true lesion pixels (clinically critical)

✓ 8-way TTA + threshold optimization without any additional training

✓ Full pipeline: X-ray in → Magseed coordinate out (mm-level accuracy)

Dataset: CBIS-DDSM · Architecture: ResUNet + Attention · Framework: PyTorch

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